

Figure 4: LLMs’ ratings on the applicability of spatial relations in both congruent and incongruent cases. SOTA models, namely GPT-4 and ZhipuAI, significantly provided higher ratings for spatial descriptions that were congruent with the ground truth, whereas the performance of the other three models was comparatively weaker, likely due to their insensitivity to spatial relations. The error bars represent the standard error of the mean (SEM). The temperature setting underlying the results presented here is 0, with the other two cases (i.e. 0.5 and 1) detailed in the Appendix Fig.S3.

3*7 subgrid) and those in the other three subgrids. As shown in Fig. 4, GPT-4 and ZhipuAI exhibit strong performance in rating the applicability of spatial descriptions, as they can effectively distinguish between descriptions that are congruent or incongruent with the actual spatial relationships. In contrast, the other three models—GPT-3.5-Turbo, Qwen-Turbo, and Llama3-8B—show significant insensitivity to spatial relations.

The results reveal that GPT-4 performs remarkably well on three types of spatial relations—namely "above", "below", and "left". However, this performance does not extend to the "right" relation, where its accuracy drops 5. Similarly, ZhipuAI also provides relatively accurate ratings for the "above" and "below" relations. Qwen-Turbo shows partial success, particularly when the "above" relation is used to describe spatial relationships between a figure object situated in the upper locations and the reference object. Other models, including GPT-3.5-Turbo and Llama3-8B, exhibit significant weaknesses in representing almost all spatial relations. Interestingly, even models that perform well in recognizing basic spatial relations still show some overlap in representing adjacent spatial relations, often spreading their ratings around the vertex of the 7*7 grid. Specifically, GPT-4’s ratings for the appropriateness of "below" descriptions are higher in the bottom-left area rather than exclusively in the bottom area, and a similar pattern is observed in ZhipuAI’s performance.

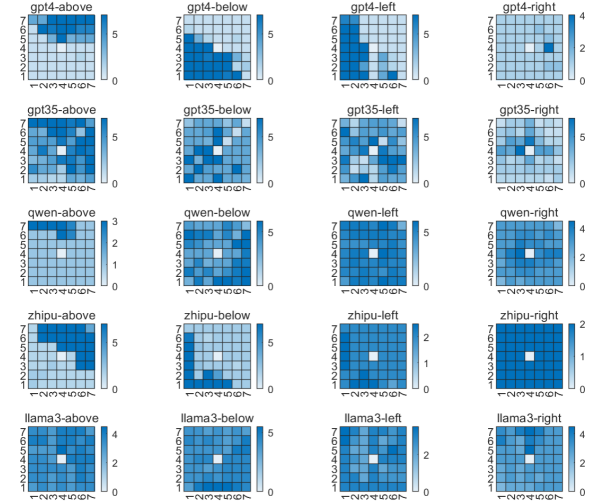
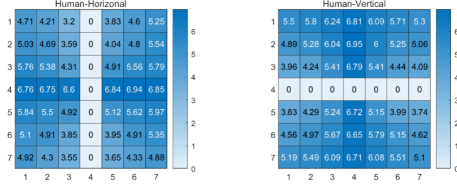


Figure 5: Performance of five LLMs on the spatial rating task. Four types of spatial relations are involved in the rating process, namely "above", "below", "left", and "right". The intensity of color bars represents models’ evaluation of the appropriateness of the spatial statements given to them. Ratings range from 1 to 7, where higher scores indicate better applicability. Temperature underlying the results shown here is 0, leaving the other two cases (i.e. 0.5 and 1) available in the Appendix Fig.S4.

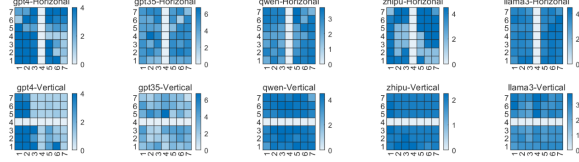
LLMs’ performance in rating the four types of spatial relations ("above", "below", "left", and "right") is averaged across horizontal and vertical directions. Specifically, the "above" and "below" relations are combined as representing the vertical axis, while the "left" and "right" relations are categorized under the horizontal axis. The resulting rating matrix is then compared with human ratings from a previous study (Hayward and Tarr, 1995). Human ratings exhibit a clear axial pattern, with ratings highest when the figure object and the reference object are aligned on the same axis, gradually decreasing as the figure object moves away from the central axis (Fig.6(a)). However, this axial pattern is not observed in any of the LLMs’ performance (Fig.6(b)).

5 Discussion

LLMs’ spatial representation abilities are evaluated through two tasks adapted from cognitive psychology: the spatial generation task and the spatial rating task, which test the models’ capacity to describe and judge spatial relationships on a 2D scene. The observed directional imbalance in the spatial generation task mirrors human tendencies (Munnich et al., 2001; Hayward and Tarr, 1995),



(a) Humans' ratings.



(b) Five LLMs' ratings with the temperature set to 0, leaving the other two cases (0.5 and 1) available in the Appendix Fig.S5.

Figure 6: Rating performance of Humans and LLMs on each location where the figure object is situated at around the reference object, averaged across horizontal and vertical directions respectively.

where vertical prepositions like "above" and "below" are used more often than horizontal ones. The lower frequency of horizontal terms suggests that LLMs' spatial depictions along the horizontal axis are coarser. This pattern is likely rooted in the effect of gravity on human daily life (Stahn et al., 2020; Lacquaniti et al., 2015; Levinson, 1996), where vertical terms tend to be more prevalent than their horizontal counterparts. Consequently, LLMs are indirectly shaped by this bias through human-oriented language.

In terms of heterogeneity in LLMs' behavior, more advanced models appear to be significantly more proficient in spatial representations. Specifically, SOTA models such as GPT-4 demonstrate greater accuracy in judging spatial relationships between objects and exhibit higher geometric richness in their choice of spatial prepositions when generating spatial descriptions compared to GPT-3.5-Turbo and Llama3-8B. This finding suggests that spatial representations can indeed be derived from spatial language, and LLMs with superior overall performance are more likely to possess enhanced spatial abilities. However, even the best-performing LLMs still fall short of perfection, indicating the need for further precision in practical applications. Additional pretraining with automatically generated spatial datasets could potentially improve LLMs' spatial reasoning (Mirzaee et al., 2021).

The influence of temperature on LLMs' perfor-

mance in both spatial tasks appears minimal, as no significant differences are observed in models' choice of spatial terms or their judgment of spatial relationships under different temperature levels (0, 0.5, and 1). Since temperature controls the randomness of model responses (Zhu et al., 2024), the insensitivity to temperature variations in spatial tasks may suggest the fundamental constancy of spatial cognition in human life. This finding aligns with studies indicating that changes in temperature have little effect on LLMs' problem-solving performance (Renze and Guven, 2024). Interestingly, all LLMs, including SOTA models like GPT-4 and ZhipuAI, fail to accurately represent rightward spatial relationships, highlighting a bias in the models' training datasets, where leftward relationships seem to be more prevalent in natural language. This phenomenon, to our knowledge, is being reported for the first time and warrants further investigation. One possible explanation is that, given most people are right-handed, leftward spatial relationships may be more intuitive and commonly used in practice.

It is also worth noting that LLMs fail to capture certain subtle characteristics of human spatial representations, such as axial salience. Cognitive psychology research has shown that humans tend to exhibit more accurate spatial representations in regions near the central axis (Hayward and Tarr, 1995), with accuracy decreasing as the distance from the axis increases. However, this tendency is absent in LLMs' performance, highlighting the limitations of models that excel at detecting regularities and generating words linearly, yet struggle with visualizing situations in a 2D space. This suggests that spatial language does not equate to spatial representation, and there may be an upper limit to the spatial representation capabilities of linguistic models.

6 Conclusion

Both similarity and difference exist between spatial representations of LLMs and humans. On one hand, LLMs resemble humans in the choice of spatial prepositions while describing spatial relationships between two objects on a 2D scene. Vertically oriented spatial terms are preferred by LLMs relative to horizontal terms, which is consistent to humans' performance and probably the reflection of gravity. On the other hand, finer representations along axis in humans do not appear in LLMs' spatial cognition, indicating that LLMs actually fail to capture

some subtle facets in human language.

Limitations

One limitation of this study is the simplification of the spatial tasks, which may not fully capture the intricate and multifaceted nature of human spatial cognition. While the tasks provide valuable insights into LLMs' spatial reasoning, they may not account for the complex, dynamic, and context-dependent factors that influence human spatial processing. Additionally, the evaluation of LLMs' spatial representations is based on textual input, which inherently may not capture the full range of spatial nuances that could be conveyed through visual input. Visual representations are known to play a crucial role in human spatial reasoning, and relying solely on text may limit the models' ability to develop a truly rich spatial understanding. Moreover, this study does not consider the potential impact of other hyperparameters—such as model architecture, training data, and optimization strategies—on LLMs' spatial performance. The tuning of these hyperparameters could influence the models' ability to generalize across different spatial tasks and scenarios.

Future research should aim to investigate LLMs' spatial representations in more complex, real-world scenarios that more closely mirror human cognition, and use a broader set of evaluation metrics that encompass both quantitative and qualitative measures. This will enable a more nuanced understanding of the models' spatial reasoning abilities. Furthermore, it would be valuable to explore techniques to enhance LLMs' spatial representations, such as the use of effective prompting strategies, incorporating multimodal inputs (e.g., images or videos), or leveraging multi-agent collaboration. These approaches could potentially mitigate current limitations and enable LLMs to achieve more sophisticated, human-like spatial reasoning.

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